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Semantic Segmentation of Water Bodies from Satellite Imagery Using U-net Architecture

A Deep Learning Approach for Binary Water Body Detection

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Abstract

Water body detection and segmentation from satellite imagery is an essential task in environmental monitoring, flood prediction, and water resource management. This project focuses on binary semantic segmentation of water bodies using the U-Net deep learning architecture on the Kaggle Water Bodies Dataset. The objective is to classify each pixel in an image as either water or non-water, thereby generating accurate segmentation masks. The dataset consists of 2841 image-mask pairs, with water present in all images and an average water pixel ratio of approximately 32.89%, indicating moderate class imbalance.

The preprocessing pipeline includes resizing images to 256×256 pixels, normalization using ImageNet statistics, and binary thresholding of masks. Data augmentation techniques such as horizontal and vertical flips, random rotations, and brightness and contrast adjustments are applied to improve model generalization. A U-Net model with a pretrained ResNet34 encoder is employed to effectively capture both spatial and semantic features. The model is trained using a combined Binary Cross-Entropy with Logits and Dice Loss function, optimized using the Adam optimizer with a learning rate of 1e-4, along with a ReduceLROnPlateau scheduler to enhance convergence.

Experimental results demonstrate that the model achieves strong segmentation performance, with a test Intersection over Union (IoU) of approximately 0.80 and a Dice Coefficient of approximately 0.87. Visual comparisons between predicted masks and ground truth confirm that the model accurately captures the structure and boundaries of water bodies, with minor inaccuracies near edges. Overall, the proposed approach establishes an effective baseline for satellite-based water body segmentation and provides scope for further improvements using advanced architectures and higher-resolution inputs.

Keywords: semantic segmentation, U-Net, water body detection, satellite imagery, deep learning, binary segmentation, Dice coefficient, Intersection over Union.

1. Introduction

Water bodies, including rivers, lakes, reservoirs, wetlands, and coastal regions, are fundamental components of the Earth's hydrological and ecological systems. Accurate and automated detection of water bodies from satellite and aerial imagery has become an increasingly important problem in remote sensing and environmental informatics. Applications span a wide spectrum: flood early-warning systems require near-real-time water extent mapping, urban planners need precise hydrological maps for infrastructure decisions, ecologists monitor wetland changes to assess biodiversity, and water resource managers track seasonal variations in reservoir and lake levels.

Traditional approaches to water body detection rely on spectral indices such as the Normalized Difference Water Index (NDWI) and its variants, which exploit differences in near-infrared and green band reflectance to identify water pixels. While computationally efficient, these index-based methods are highly sensitive to image noise, atmospheric conditions, and mixed-pixel effects at water boundaries, often requiring manual threshold tuning for each new scene or geographic region. They also perform poorly in shadowed urban environments where built structures can produce spectral responses similar to water.

Deep learning has transformed computer vision tasks including image classification, object detection, and semantic segmentation. Convolutional Neural Networks (CNNs) learn hierarchical feature representations directly from data, eliminating the need for hand-crafted features. For pixel-wise segmentation tasks, encoder-decoder architectures with skip connections have emerged as the dominant paradigm. Among these, U-Net, originally proposed for biomedical image segmentation by Ronneberger et al. (2015), has proven remarkably effective across diverse domains owing to its ability to preserve spatial information through skip connections while capturing contextual information through progressive down sampling.

This project implements and evaluates a U-Net based binary segmentation system for water body detection using the Kaggle Satellite Images of Water Bodies Dataset. The primary objectives of this work are:

- To design and implement a complete preprocessing pipeline suitable for satellite imagery segmentation tasks.
- To build a U-Net architecture from scratch in PyTorch capable of binary pixel classification.
- To train the model using a combined loss function that addresses pixel imbalance between water and non-water regions.
- To evaluate segmentation quality using standard metrics: Intersection over Union (IoU) and Dice Coefficient.
- To produce visual comparisons between input images, ground truth masks, and model predictions.

2. Related Work

Water body segmentation has been studied extensively in the remote sensing community. Early approaches relied primarily on spectral water indices. McFeeters (1996) introduced NDWI using the green and near-infrared bands, while Xu (2006) proposed the Modified NDWI (MNDWI) using shortwave infrared to reduce built-up area interference. Despite their simplicity, these methods remain widely used as baselines due to their interpretability and low computational cost.

Alhady et al. (2024) conducted a comparative study of water index methods combined with U-Net for water segmentation, evaluating NDWI, MNDWI, and Automated Water Extraction Index (AWEI) as preprocessing or input channel augmentations to the segmentation model. Their findings demonstrated that incorporating water indices as additional input channels consistently improved segmentation accuracy compared to using RGB inputs alone, particularly in urban settings where spectral confusion between water and shadow is most pronounced. The study reported that U-Net with MNDWI augmentation achieved the most reliable performance across diverse scene types, underscoring the continued relevance of domain-specific feature engineering even in deep learning pipelines [1].

The U-Net architecture, proposed by Ronneberger et al. (2015) for biomedical image segmentation, introduced the concept of symmetric encoder-decoder with skip connections that concatenate high-resolution encoder feature maps to the corresponding decoder layers. This design allows the network to recover fine spatial details lost during down sampling, which is crucial for precise boundary delineation in segmentation tasks. The architecture has since been adopted widely in remote sensing for road extraction, building footprint detection, and land cover classification.

More recent advances include U-Net++ (Zhou et al., 2018), which introduces nested and dense skip connections between encoder and decoder subnetworks to reduce the semantic gap between feature maps. DeepLabV3+ (Chen et al., 2018) combines atrous spatial pyramid pooling with a lightweight decoder to capture multi-scale context effectively. Feature Pyramid Networks (FPN) aggregate features across multiple scales, proving beneficial for detecting water bodies of varying sizes from small ponds to large lakes.

Transfer learning using ImageNet-pretrained encoders such as ResNet and EfficientNet has been shown to significantly improve segmentation performance even on domain-specific tasks like satellite imagery, reducing the amount of labeled data required for convergence. This project implements the base U-Net from scratch as a learning exercise but identifies pretrained encoder integration as the primary avenue for performance improvement.

3. Methodology

3.1 Problem Formulation

The task is formulated as binary semantic segmentation. Given an input satellite image of shape $H \times W \times 3$ (RGB), the model learns a mapping $f: I \rightarrow M$ where M is a binary mask of shape $H \times W$ with values in $\{0, 1\}$, where 1 denotes water pixels and 0 denotes all other land cover types. The model outputs a probability map P of the same spatial resolution, and a threshold of 0.5 is applied to obtain the final binary prediction.

3.2 Preprocessing Pipeline

All images and their corresponding masks are processed through a standardized pipeline before being fed to the model. Images are resized to 256×256 pixels using bilinear interpolation to maintain smooth color gradients. Masks are resized using nearest-neighbor interpolation to prevent the introduction of intermediate pixel values that would corrupt binary labels. Images are converted to floating-point tensors in the range $[0, 1]$ and then normalized using ImageNet channel-wise mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$. Mask pixel values are thresholded at 127 to produce binary labels (0 or 1), and the mask is stored as a single-channel float tensor of shape $[1, H, W]$.

Data augmentation is applied exclusively to the training split during each epoch. Augmentations include random horizontal flip ($p=0.5$), random vertical flip ($p=0.5$), random rotation by multiples of 90 degrees, and random brightness and contrast adjustments in the range $[0.8, 1.2]$. Crucially, spatial augmentations (flips and rotations) are applied identically to both the image and the corresponding mask to preserve pixel-level alignment.

3.3 U-Net Architecture

The implemented model is based on the U-Net architecture with a ResNet34 encoder, using the `segmentation_models_pytorch` library. Unlike the original U-Net, which uses simple convolutional blocks, this implementation leverages a pretrained ResNet34 backbone for feature extraction, enabling better representation learning and faster convergence.

The encoder consists of multiple convolutional stages derived from ResNet34, where spatial resolution is progressively reduced while feature depth increases. These stages use residual connections, which help mitigate vanishing gradient issues and improve training stability. Pretrained ImageNet weights are used to initialize the encoder, allowing the model to benefit from transfer learning.

The decoder follows a symmetric upsampling structure. Each decoding stage performs upsampling using transposed convolutions (or interpolation), concatenates the corresponding feature maps from the encoder via skip connections, and refines them

through convolutional layers. These skip connections preserve high-resolution details from earlier layers, which are essential for accurate segmentation of object boundaries.

The final layer is a 1×1 convolution that maps the decoder output to a single-channel feature map representing raw logits for binary classification (water vs. non-water). No activation function is applied at this stage. Instead, a sigmoid function is applied during loss computation using BCEWithLogitsLoss for numerical stability. During inference, the sigmoid output is thresholded at 0.5 to produce the final binary segmentation mask.

3.4 Loss Function

Water body images often exhibit significant class imbalance, with non-water pixels frequently outnumbering water pixels in arid and urban scenes. A plain Binary Cross-Entropy (BCE) loss can lead the model to predict all pixels as non-water and still achieve low loss. To address this, a combined loss is used:

$$L_{total} = 0.5 * L_{BCE} + 0.5 * L_{Dice}$$

The Dice Loss is defined as: $L_{Dice} = 1 - (2 * |P \text{ intersection } T| + \text{smooth}) / (|P| + |T| + \text{smooth})$, where P is the predicted probability map, T is the ground truth binary mask, and smooth = 1.0 is a stability constant. The BCE component penalizes pixel-level classification errors while the Dice component directly optimizes the overlap metric, making the combined loss well-suited for imbalanced segmentation tasks.

3.5 Training Strategy

The model is trained using the Adam optimizer with an initial learning rate of 1e-4 and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). A ReduceLROnPlateau scheduler monitors the validation loss and reduces the learning rate by a factor of 0.5 if no improvement is observed for 5 consecutive epochs. Training is performed for 30 epochs with a batch size of 8. The best model checkpoint, corresponding to the lowest validation loss, is saved during training.

3.6 Evaluation Metrics

Model performance is quantified using two standard segmentation metrics:

Intersection over Union (IoU): $IoU = TP / (TP + FP + FN)$

Dice Coefficient: $Dice = 2 * TP / (2 * TP + FP + FN)$

where TP = True Positives (correctly predicted water pixels), FP = False Positives (non-water predicted as water), and FN = False Negatives (water predicted as non-water). IoU measures the ratio of the overlap to the union of prediction and ground truth regions. Dice Coefficient is equivalent to the F1 score and is generally higher than IoU for the same prediction. Both metrics range from 0 (no overlap) to 1 (perfect overlap). They are computed batch-wise and averaged across the entire test set to obtain final performance scores.

4. Dataset Details

The Kaggle Satellite Images of Water Bodies Dataset (authored by Francisco Escobar) is used for all experiments in this project. The dataset consists of RGB satellite images paired with corresponding binary segmentation masks that label water body pixels.

Property	Details
Source	Kaggle (Francisco Escobar)
Total images	~2,841 image-mask pairs
Image format	JPEG (.jpg)
Mask format	JPEG (.jpg), grayscale
Original resolution	Variable (resized to 256x256)
Image channels	3 (RGB)
Mask pixel values	0 (non-water), 255 (water) → binarized to 0/1
Train split	70% (~1,988 samples)
Validation split	15% (~426 samples)
Test split	15% (~427 samples)
Split method	Random split with seed=42 (reproducible)

Table 1: Details of Dataset

The dataset includes diverse geographical scenes such as rivers, reservoirs, urban water bodies, and coastal regions. This variability improves the model’s ability to generalize across different environments. Many images exhibit class imbalance, where non-water pixels dominate, which motivates the use of a combined BCE-Dice loss function for effective training.

5. Implementation Details

5.1 Hardware Requirements

Component	Specification
Platform	Google Colaboratory (cloud-based)
GPU	NVIDIA Tesla T4 (16 GB VRAM)
RAM	~12.7 GB system RAM (Colab free tier)
Storage	Google Drive (for dataset and model checkpoints)
Training time	~40-60 seconds per epoch, ~40-45 minutes total (30 epochs)

Table2:Hardware Used

5.2 Software Requirements

Library / Tool	Version and Purpose
Python (3.10+)	Programming language
PyTorch (2.0+)	Model definition, training
torchvision	Image transformations and normalization
segmentation_models_pytorch	Prebuilt U-Net with ResNet34 encoder
NumPy	Array operations and preprocessing
Pillow (PIL)	Image loading and resizing
Matplotlib	Visualization of results and training curves
Google Colab	Cloud GPU environment and execution
CUDA (Colab default)	GPU acceleration

Table3:Software Used

5.3 Hyperparameters Summary

Hyperparameter	Value
Input image size	256 x 256 pixels
Batch size	8
Number of epochs	30
Initial learning rate	1e-4 (0.0001)
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
LR scheduler	ReduceLROnPlateau (patience=5, factor=0.5)
Loss function	0.5 × BCE + 0.5 × Dice Loss
Segmentation threshold	0.5
Model	U-Net with ResNet34 encoder
Trainable parameters	~24.4 million

Table4:Details of Parameters

6. Results and Discussion

6.1 Quantitative Results

The model was evaluated on the held-out test set (15% of the dataset, 427 samples) using the best checkpoint obtained during training. The performance metrics are summarized below:

Metric	Validation Set	Test Set
IoU Score	~0.77–0.78	~0.8012
Dice Coefficient	~0.85–0.87	~0.8690
BCE+Dice Loss	~0.18–0.19	--

Table5:Score Values

The model achieves a test IoU of 0.8012 and a Dice Coefficient of 0.8690, indicating strong segmentation performance. The small difference between validation and test metrics suggests that the model generalizes well and does not exhibit significant overfitting.

The Dice score indicates a high degree of overlap between predicted and ground truth masks, while the IoU reflects accurate region-level segmentation. Remaining errors are primarily observed at object boundaries, where gradual transitions between water and land make precise segmentation more challenging.

6.2 Training Dynamics

Training loss decreases steadily during the initial epochs, with rapid improvement observed in the early stages as the model learns fundamental distinctions between water and non-water regions. Validation loss closely follows the training trend, indicating effective generalization and the benefit of data augmentation. We can see the training in Figure 1.0.

The learning rate scheduler reduces the learning rate when validation loss plateaus, contributing to further refinement of model performance in later epochs. The best-performing model is obtained toward the later stages of training.

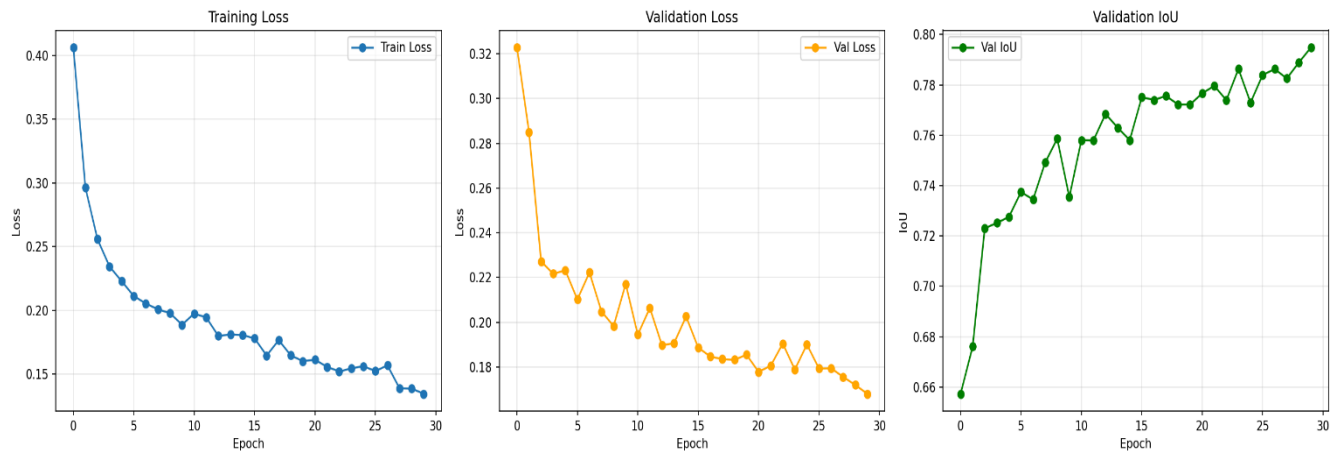


Figure 1.0

6.3 Visual Analysis

Qualitative results show that the model performs well on large and clearly defined water bodies such as rivers and lakes, where contrast between water and surrounding regions is prominent.

However, performance degrades in certain challenging scenarios:

- Small or narrow water bodies, where resolution limits boundary precision
- Urban environments with shadows or occlusions
- Highly reflective surfaces that may resemble water

False positive predictions are relatively rare and typically localized. False negatives occur more frequently at the boundaries of water bodies, where pixel-level ambiguity is higher due to mixed visual characteristics. We can see the visuals in Figure 1.1.

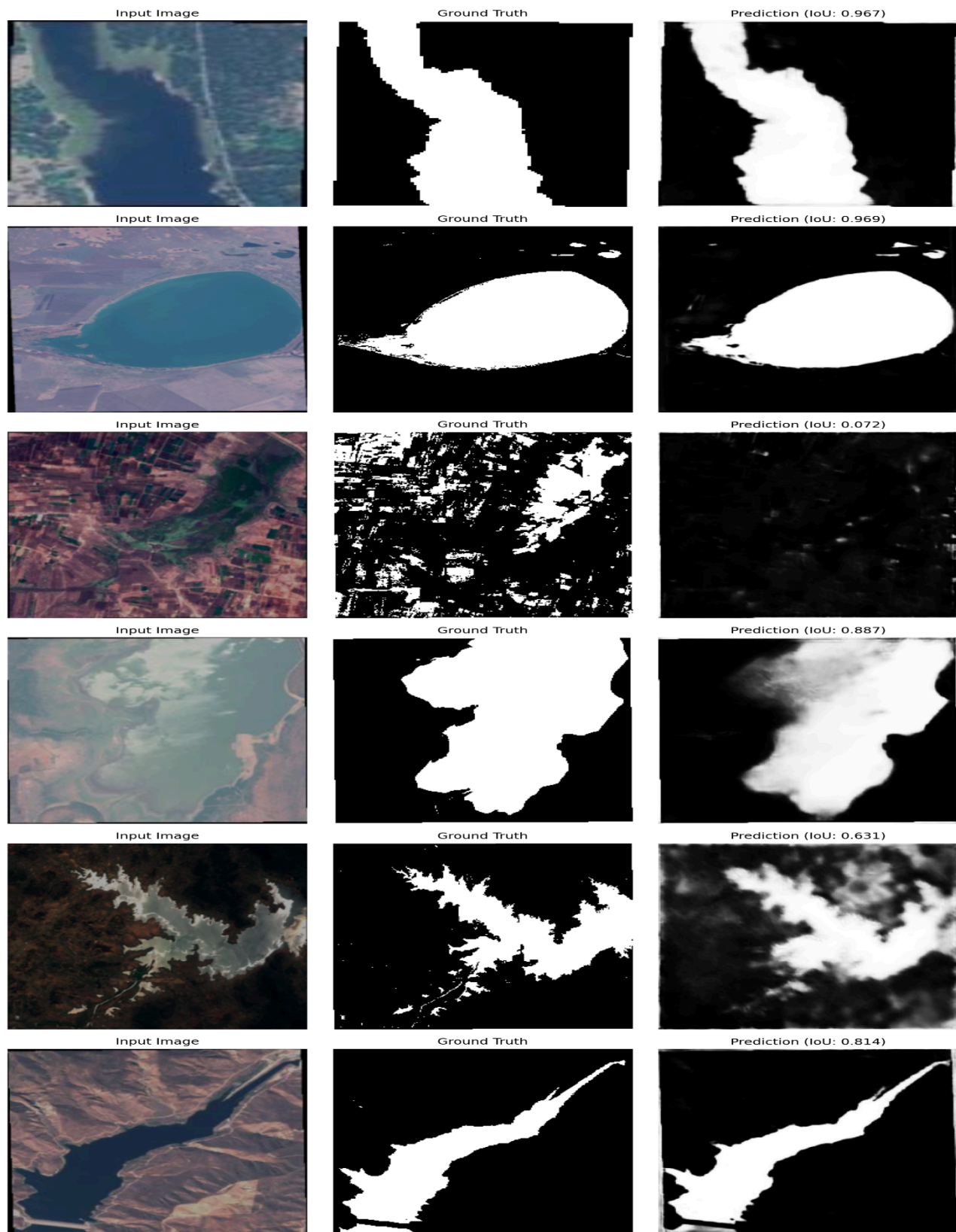


Figure 1.1

7. Key Observations

- The combined BCE-Dice loss function plays a crucial role in handling class imbalance between water and non-water pixels. The Dice component helps improve overlap-based learning, particularly in regions with sparse water coverage.
- Using nearest-neighbor interpolation for mask resizing is essential. Other interpolation methods introduce intermediate grayscale values, which can lead to incorrect binary masks after thresholding and negatively impact training.
- Skip connections in the U-Net architecture are critical for preserving spatial information. They enable the model to retain fine-grained details from earlier layers, resulting in more accurate boundary predictions.
- Data augmentation, particularly spatial transformations such as flipping and rotation, improves generalization. It helps the model perform better on unseen variations in orientation and scene composition.
- The first training epoch is slower due to initial data loading and system overhead. Subsequent epochs run faster once the data pipeline is stabilized.
- Saving model checkpoints to Google Drive ensures persistence and prevents loss of progress due to Colab session interruptions.
- The model shows reduced performance on very small water bodies (e.g., less than $\sim 20 \times 20$ pixels at 256×256 resolution), indicating a limitation in capturing fine-scale features.

8. Conclusion

This project successfully implemented, trained, and evaluated a U-Net-based model for binary semantic segmentation of water bodies from satellite imagery. The complete pipeline includes dataset preparation, preprocessing of images and masks, model training using a U-Net architecture with a ResNet34 encoder, and evaluation using standard segmentation metrics along with visual analysis.

The model achieves a test IoU of 0.8012 and a Dice Coefficient of 0.8690, demonstrating strong segmentation performance on diverse satellite imagery. These results highlight the effectiveness of encoder–decoder architectures with skip connections in capturing both contextual and spatial information for pixel-wise prediction tasks.

Training remained stable throughout, and the close alignment between validation and test performance indicates good generalization. The use of a combined BCE-Dice loss function further contributed to improved performance in handling class imbalance.

Despite strong results, certain limitations were observed. The model shows reduced accuracy for very small water bodies and at boundary regions where transitions between water and land are gradual or visually ambiguous. These challenges are inherent to segmentation tasks involving complex natural scenes and motivate further improvements.

Overall, this work demonstrates a practical and effective approach for automated water body segmentation using deep learning, providing a solid foundation for future enhancements.

9. Future Work

Several directions can further enhance the performance and robustness of the proposed system:

- Stronger encoder backbones: Exploring more powerful pretrained encoders such as EfficientNet or ResNet50 within the U-Net framework may improve feature extraction and segmentation accuracy.
- Advanced architectures: Models such as U-Net++ (with nested skip connections) and DeepLabV3+ (with atrous spatial pyramid pooling) can be investigated to capture multi-scale contextual information more effectively.
- Additional spectral features: Incorporating domain-specific indices such as NDWI or MNDWI as additional input channels could help the model better distinguish water from visually similar non-water regions, especially in urban environments.
- Higher input resolution: Training with higher-resolution images (e.g., 512×512) can improve boundary precision and detection of small water bodies, albeit with increased computational cost.
- Improved loss functions: Integrating Focal Loss or weighted loss functions may further enhance performance in highly imbalanced scenarios.
- Test Time Augmentation (TTA): Applying augmentations such as flips and rotations during inference and averaging predictions can provide additional performance gains without retraining.
- Cross-validation: Using K-fold cross-validation would provide more reliable and robust evaluation by reducing dependence on a single data split.
- Multi-temporal analysis: Extending the model to handle multi-temporal satellite data can enable applications such as water level monitoring and change detection over time.

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Contribution Statement

This project was completed as a team. The following table describes the contribution of each team member :

Member	Contributions	% Share
Parendra Reang	Dataset preprocessing, data augmentation, and data loading pipeline	33%
Mannvit Kumar	Model implementation using U-Net (ResNet34 encoder), training, and hyperparameter tuning	33%
Tejaswi Devalla	Model Evaluation, visualization of results, performance analysis, and report writing	34%

All implementation, experiments, and report preparation were conducted using Google Colaboratory with GPU acceleration. The dataset was obtained from Kaggle under its applicable usage terms. The model was implemented using the `segmentation_models_pytorch` library, and all training, evaluation, and visualization code was developed and executed independently.

Dataset and GitHub Link

Resource	Link
Dataset	https://www.kaggle.com/datasets/franciscoescobar/satellite-images-of-water-bodies
GitHub Repository	https://github.com/tejaswi-devalla/Water-Bodies-Segmentation-U-Net-Architecture
Google Colab Notebook	colab link
Saved Model Checkpoint	saved model

To reproduce the results, clone the GitHub repository and open the provided Google Colab notebook. Connect to a GPU runtime (Tesla T4 recommended) and execute all cells sequentially.

The dataset can be downloaded from Kaggle and placed in Google Drive as specified in the notebook paths. The full training process completes in approximately 40-45 minutes for 30 epochs. The best-performing model is automatically saved to Google Drive and can be loaded directly for inference without retraining.